MAHATMA GANDHI UNIVERSITY, KOTTAYAM
SECOND SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION
April 2025

Discipline Specific Core Course - MG2DSCECC102 – Fundamentals of OS and Linux

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours

Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Scheme of Evaluation

Part A

Very Short Answer Questions

Each Question carries one mark

Answer all questions.

1. **Name the technological innovation which led to the development of time-sharing systems? [K] [CO1]**
   Multiprogramming

2. **How the status of the current activity of a process can be identified? [U][C02]**
   Program Counter

3. **A system allocates memory in fixed-size blocks, allowing processes to be stored in non-contiguous locations. What is this memory management technique called? [K][CO3]**
   Paging

4. **What way an Operating System helps the computer? [U][CO1]**
   Acts as an interface between the user and the hardware

5. **Priority Scheduling led to starvation. Why? [U] [CO2]**
   Low-priority processes get indefinitely delayed due to the continuous arrival of higher-priority processes.

6. **Two processes require 200MB each. The system has free blocks of 150MB, 250MB, and 300 MB. One process is allocated in the 250MB block. Name the fragmentation? [U][CO3]**
   External fragmentation

7. **Which OS structure stores information about a process? [K] [CO2]**
   Process Control Block

8. **Why is a segment table needed in segmentation? [U][CO3]**
   It maps logical addresses to physical addresses and helps manage memory efficiently.

9. **Yes or No: A process moves directly from the 'new' state to the 'running' state. [K] [CO2]**
   No

10. **How does the system interpret an object file? [U][CO3]**
    Binary Format, Symbols, and Relocations

(10 × 1 = 10)

Part B

Short Answer Questions
Answer any 4 questions.
Each question carries 5 marks

11. Identify the different system calls for device management. **[U] [C01]**

Introduction – 2 marks
Any 3 or more – 3 marks

12. What are the differences between CPU-bound and I/O- bound processes? **[K] [CO2]**

One mark for each comparison

13. How does the acyclic graph directory improve file sharing compared to a tree-structured directory? **[U][CO3]**

Introduction- 2 marks
Comparison – 3 marks

14. Explain how processes move between different scheduling queues. **[U] [CO2]**

Scheduling queue introduction – 2 marks
Explanation -3 marks

15. Explain why linked allocation is better suited for files that frequently grow in size. **[U] [CO3]**

Linked allocation introduction -2 marks
Explanation – 3 marks

16. Identify the operating system services for the system. **[U][CO1]**

One mark for each service

(4 × 5 = 20)

Part C

Essay Questions
Answer any 2 questions.
Each question carries 10 marks

17. Identify the different types of operating systems. **[U][CO1]**

OS definition -2 marks
Only three types – 5 marks
Or
More than three types – 8 marks

18. Explain the concept of a Resource Allocation Graph in operating systems. Provide an example to illustrate the detection of deadlock using Resource Allocation Graph. **[U] [CO2]**

Introduction – 5 marks
Figure with markings- 2 marks
Example to detect deadlock with figure -3 marks

19. What is the purpose of a page replacement algorithm? List the different types of page replacement algorithms. **[K][CO3]**

Page replacement algorithm introduction – 2.5 marks
For each type with examples – 2.5 marks

(10 x 2 =20)